In [22]:
```
from sklearn.naive_bayes import CategoricalNB
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import KBinsDiscretizer

credit = pd.read_csv('credit.csv')

kbins = KBinsDiscretizer(n_bins=4, encode='ordinal', strategy='quantile')

credit['months_loan_duration_bins'] = kbins.fit_transform(credit[['months_loan_duration']])
credit[['months_duration_0', 'months_duration_1','months_duration_2', 'months_duration_3']] = pd.get_dummies(credit['

credit['amount_bins'] = kbins.fit_transform(credit[['amount']])
credit[['amount_0', 'amount_1', 'amount_2', 'amount_3']] = pd.get_dummies(credit["amount_bins"])

credit['age_bins'] = kbins.fit_transform(credit[['age']])
credit[['age_0', 'age_1', 'age_2', 'age_3']] = pd.get_dummies(credit["age_bins"])
```

```
C:\Users\Bushinii\anaconda3\Lib\site-packages\sklearn\preprocessing\_discretization.py:304: FutureWarning: The curren
t default behavior, quantile_method='linear', will be changed to quantile_method='averaged_inverted_cdf' in scikit-le
arn version 1.9 to naturally support sample weight equivalence properties by default. Pass quantile_method='averaged_
inverted_cdf' explicitly to silence this warning.
  warnings.warn(
C:\Users\Bushinii\anaconda3\Lib\site-packages\sklearn\preprocessing\_discretization.py:304: FutureWarning: The curren
t default behavior, quantile_method='linear', will be changed to quantile_method='averaged_inverted_cdf' in scikit-le
arn version 1.9 to naturally support sample weight equivalence properties by default. Pass quantile_method='averaged_
inverted_cdf' explicitly to silence this warning.
  warnings.warn(
C:\Users\Bushinii\anaconda3\Lib\site-packages\sklearn\preprocessing\_discretization.py:304: FutureWarning: The curren
t default behavior, quantile_method='linear', will be changed to quantile_method='averaged_inverted_cdf' in scikit-le
arn version 1.9 to naturally support sample weight equivalence properties by default. Pass quantile_method='averaged_
inverted_cdf' explicitly to silence this warning.
  warnings.warn(
```

In [2]:
```
credit[['CB1_200', 'CB0', 'CBmayor200', 'CBunknown']] = pd.get_dummies(credit["checking_balance"])
credit[['CHcritical', 'CHgood', 'CHperfect', 'CHpoor', 'CHverygood']] = pd.get_dummies(credit["credit_history"])
credit[['SB100_500', 'SB500_1000', 'SEmenor100', 'SBmayor1000', 'SBunkown']] = pd.get_dummies(credit["savings_balance
credit[['ED1_4','ED4_7','EDmenor1', 'EDmayor7', "EDunemployed"]] = pd.get_dummies(credit["employment_duration"])
credit[["Hother", "Hown", "Hrent"]] = pd.get_dummies(credit["housing"])
```

```
credit[["Jmanagement", "Jskilled","Junemployed", "Junskilled"]] = pd.get_dummies(credit["job"])
credit [["Pno", "Pyes"]] = pd.get_dummies(credit["phone"])

credit["y"] = np.where(credit["default"]=="no", 0, 1)
```

In [3]:
```
num_y = credit["default"].value_counts()
min_num = min(num_y)
min_class = num_y[num_y==min(num_y)].index[0]
undersamp = credit[credit["default"]!=min_class].sample(min_num, random_state=20)
credit = pd.concat([undersamp, credit[credit["default"]==min_class]], axis=0)
credit["default"].value_counts()
```

Out[3]:
```
default
no     300
yes    300
Name: count, dtype: int64
```

In [4]:
```
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
from sklearn.metrics import confusion_matrix
import seaborn as sns

x_credit = credit.drop(['months_loan_duration', 'months_loan_duration_bins', "amount", "amount_bins", 'percent_of_inc
y_credit = credit["y"]

x_train, x_test, y_train, y_test = train_test_split(x_credit, y_credit, train_size=0.8, random_state=14)
```

In [5]:
```
from sklearn.naive_bayes import CategoricalNB

classifier = CategoricalNB()
classifier.fit(x_train, y_train)
classifier.score(x_train, y_train)
```

Out[5]:
```
0.73125
```

In [6]:
```
!pip install mlxtend
```

```
Collecting mlxtend
  Using cached mlxtend-0.24.0-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: scipy>=1.16.3 in c:\users\bushinii\anaconda3\lib\site-packages (from mlxtend) (1.16.3)
Requirement already satisfied: numpy>=2.3.5 in c:\users\bushinii\anaconda3\lib\site-packages (from mlxtend) (2.3.5)
Requirement already satisfied: pandas>=2.3.3 in c:\users\bushinii\anaconda3\lib\site-packages (from mlxtend) (2.3.3)
Requirement already satisfied: scikit-learn>=1.8.0 in c:\users\bushinii\anaconda3\lib\site-packages (from mlxtend)
(1.8.0)
Collecting matplotlib>=3.10.8 (from mlxtend)
  Using cached matplotlib-3.10.8-cp313-cp313-win_amd64.whl.metadata (52 kB)
Requirement already satisfied: joblib>=1.5.2 in c:\users\bushinii\anaconda3\lib\site-packages (from mlxtend) (1.5.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\bushinii\anaconda3\lib\site-packages (from matplotlib>=3.
10.8->mlxtend) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\bushinii\anaconda3\lib\site-packages (from matplotlib>=3.10.8
->mlxtend) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\bushinii\anaconda3\lib\site-packages (from matplotlib>=
3.10.8->mlxtend) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\bushinii\anaconda3\lib\site-packages (from matplotlib>=
3.10.8->mlxtend) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\bushinii\anaconda3\lib\site-packages (from matplotlib>=3.1
0.8->mlxtend) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\bushinii\anaconda3\lib\site-packages (from matplotlib>=3.10.8->m
lxtend) (12.0.0)
Requirement already satisfied: pyparsing>=3 in c:\users\bushinii\anaconda3\lib\site-packages (from matplotlib>=3.10.8
->mlxtend) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\bushinii\anaconda3\lib\site-packages (from matplotlib
>=3.10.8->mlxtend) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\bushinii\anaconda3\lib\site-packages (from pandas>=2.3.3->mlx
tend) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\bushinii\anaconda3\lib\site-packages (from pandas>=2.3.3->m
lxtend) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\bushinii\anaconda3\lib\site-packages (from python-dateutil>=2.7->
matplotlib>=3.10.8->mlxtend) (1.17.0)
Requirement already satisfied: threadpoolctl>=3.2.0 in c:\users\bushinii\anaconda3\lib\site-packages (from scikit-lea
rn>=1.8.0->mlxtend) (3.5.0)
Using cached mlxtend-0.24.0-py3-none-any.whl (1.4 MB)
Using cached matplotlib-3.10.8-cp313-cp313-win_amd64.whl (8.1 MB)
Installing collected packages: matplotlib, mlxtend

  Attempting uninstall: matplotlib

    Found existing installation: matplotlib 3.10.6
```

```
 Uninstalling matplotlib-3.10.6:

---------------------------------------- 0/2 [matplotlib]
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---------------------------------------- 0/2 [matplotlib]
---------------------------------------- 0/2 [matplotlib]
---------------------------------------- 0/2 [matplotlib]
---------------------------------------- 0/2 [matplotlib]
   Successfully uninstalled matplotlib-3.10.6
---------------------------------------- 0/2 [matplotlib]
---------------------------------------- 0/2 [matplotlib]
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```

```
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   ---------------------------------------- 0/2 [matplotlib]
   ---------------------------------------- 0/2 [matplotlib]
   ---------------------------------------- 0/2 [matplotlib]
   ---------------------------------------- 0/2 [matplotlib]
   ---------------------------------------- 0/2 [matplotlib]
   ---------------------------------------- 0/2 [matplotlib]
   ---------- ---------- 1/2 [mlxtend]
   ---------- ---------- 1/2 [mlxtend]
   ---------- ---------- 1/2 [mlxtend]
   ---------- ---------- 2/2 [mlxtend]

Successfully installed matplotlib-3.10.8 mlxtend-0.24.0
  WARNING: Failed to remove contents in a temporary directory 'C:\Users\Bushinii\AppData\Local\Temp\pip-uninstall-3xe
zdy6_'.
  You can safely remove it manually.
```

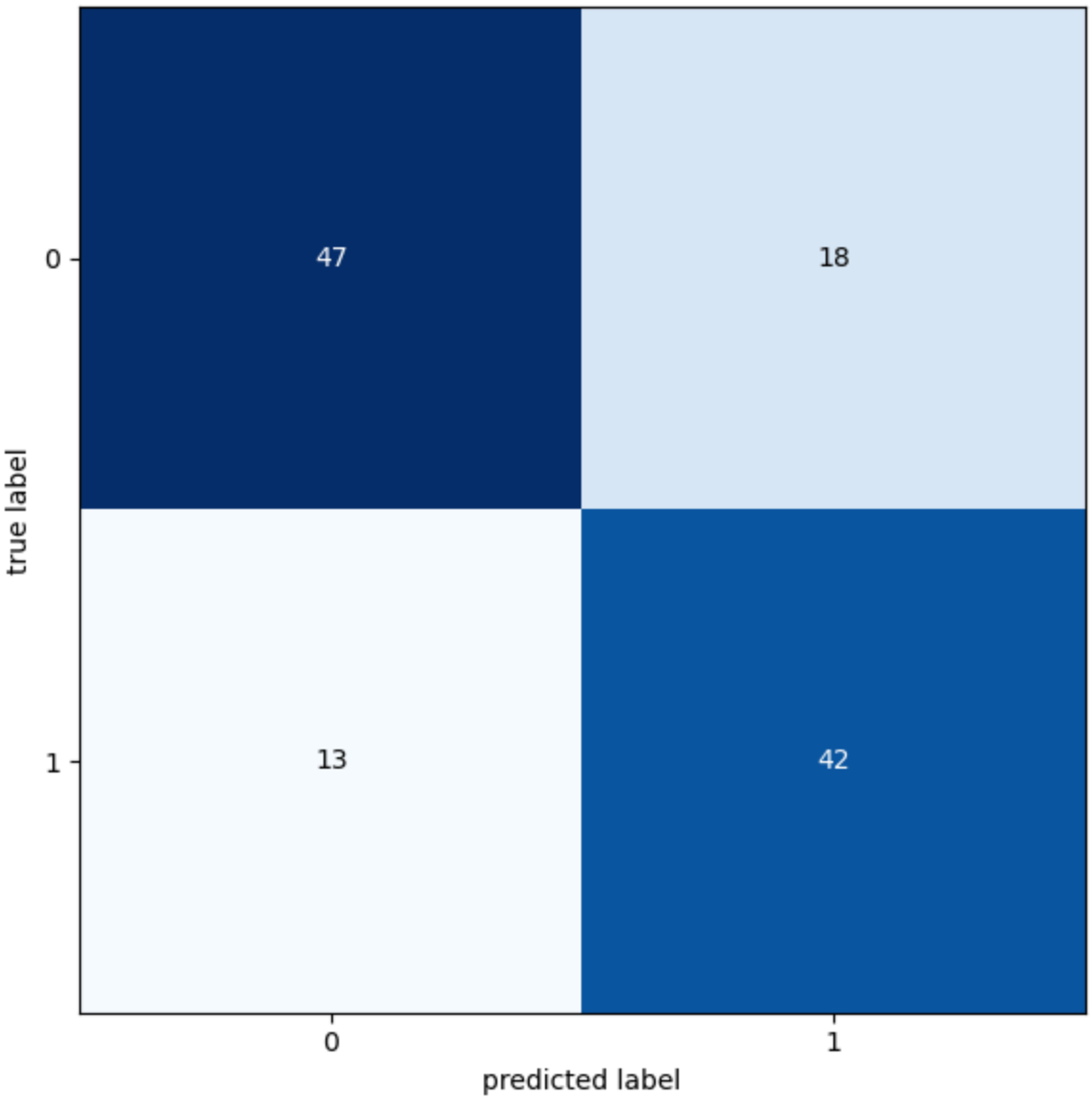
In [7]:

```python
from mlxtend.plotting import plot_confusion_matrix
y_pred = classifier.predict(x_test)

from sklearn.metrics import confusion_matrix, accuracy_score, classification_report
cm = confusion_matrix(y_test, y_pred)
plot_confusion_matrix(conf_mat=cm, figsize=(6,6), show_normed=False)
plt.tight_layout()

print(accuracy_score(y_test, y_pred))
```

```
0.7416666666666667
```



In [8]:
```
print(classification_report(y_test, round(pd.Series(classifier.predict(x_test)))))
```

```
              precision    recall  f1-score   support

           0       0.78      0.72      0.75        65
           1       0.70      0.76      0.73        55

    accuracy                           0.74       120
   macro avg       0.74      0.74      0.74       120
weighted avg       0.75      0.74      0.74       120
```

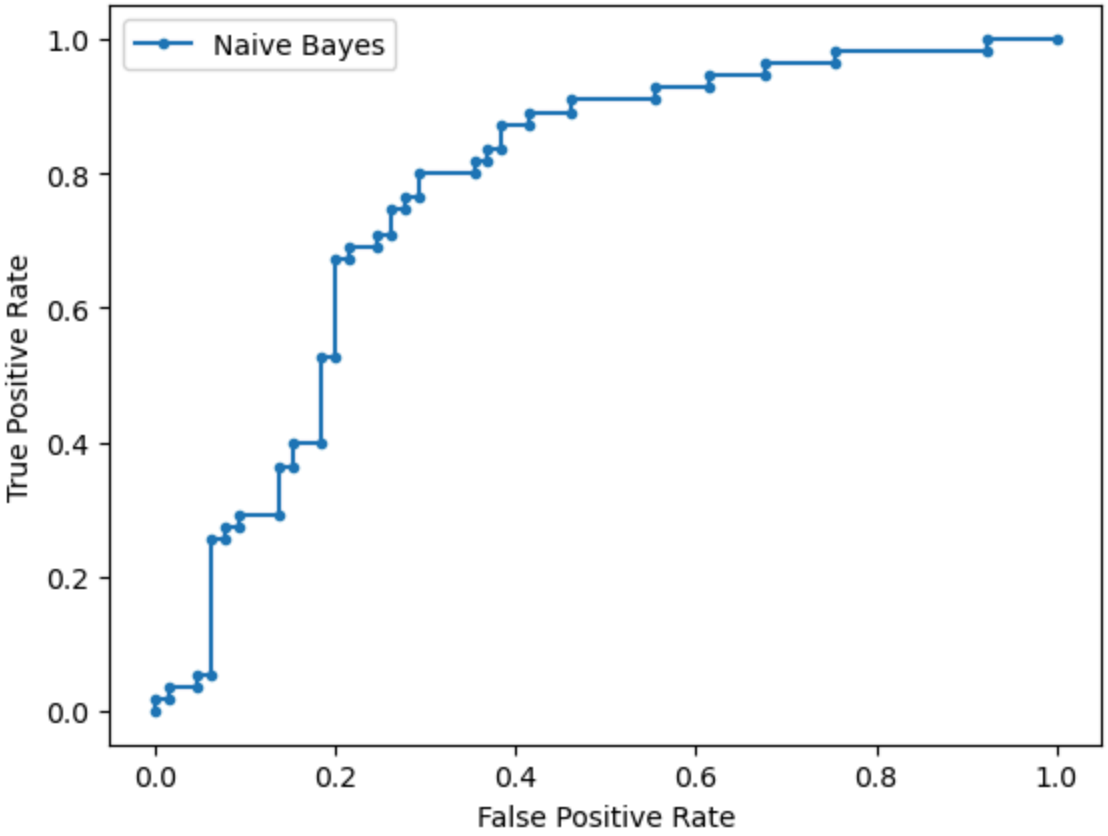
In [9]:

```python
from sklearn.metrics import roc_auc_score, roc_curve
from matplotlib import pyplot

auc = roc_auc_score(y_test, classifier.predict_proba(x_test)[:,1])
print('AUC: %.3f' % auc)

fpr, tpr, _ = roc_curve(y_test, classifier.predict_proba(x_test)[:,1])
pyplot.plot(fpr, tpr, marker='.', label='Naive Bayes') # Nota: El PDF dice 'Random Forest' en el label pero el modelo
pyplot.xlabel('False Positive Rate')
pyplot.ylabel('True Positive Rate')
pyplot.legend()
pyplot.show()
```

```
AUC: 0.775
```



In [10]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

In [12]:
```
dataset = pd.read_csv('Social_Network_Ads.csv')
X = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].values
```

In [13]:
```
dataset
```

Out[13]:

	Age	EstimatedSalary	Purchased
0	19	19000	0
1	35	20000	0
2	26	43000	0
3	27	57000	0
4	19	76000	0
...	...	...	...
395	46	41000	1
396	51	23000	1
397	50	20000	1
398	36	33000	0
399	49	36000	1

400 rows × 3 columns

In [14]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.3, random_state = 0)
```

In [15]:
```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

In [16]:
```
print(X_train)
```

```
[[-1.1631724  -1.5849703 ]
 [ 2.17018137  0.93098672]
 [ 0.0133054   1.22017719]
 [ 0.20938504  1.07558195]
 [ 0.40546467 -0.48604654]
 [-0.28081405 -0.31253226]
 [ 0.99370357 -0.8330751 ]
 [ 0.99370357  1.8563962 ]
 [ 0.0133054   1.24909623]
 [-0.86905295  2.26126285]
 [-1.1631724  -1.5849703 ]
 [ 2.17018137 -0.80415605]
 [-1.35925203 -1.46929411]
 [ 0.40546467  2.2901819 ]
 [ 0.79762394  0.75747245]
 [-0.96709276 -0.31253226]
 [ 0.11134522  0.75747245]
 [-0.96709276  0.55503912]
 [ 0.30742485  0.06341534]
 [ 0.69958412 -1.26686079]
 [-0.47689368 -0.0233418 ]
 [-1.7514113   0.3526058 ]
 [-0.67297331  0.12125343]
 [ 0.40546467  0.29476771]
 [-0.28081405  0.06341534]
 [-0.47689368  2.2901819 ]
 [ 0.20938504  0.03449629]
 [ 1.28782302  2.20342476]
 [ 0.79762394  0.26584866]
 [-0.28081405  0.15017248]
 [ 0.0133054  -0.54388463]
 [-0.18277423  0.15017248]
 [-0.08473441  0.23692961]
 [ 0.0133054  -0.25469417]
 [ 2.17018137  1.104501  ]
 [-1.7514113   0.3526058 ]
 [ 1.87606192  0.12125343]
 [ 0.40546467 -0.13901799]
 [-1.1631724   0.29476771]
 [ 0.79762394  1.36477242]
 [-0.28081405 -0.25469417]
 [-1.65337148 -0.05226085]
```

```
[-0.96709276 -0.74631796]
[ 0.30742485  0.49720103]
[-0.08473441 -1.06442747]
[-1.06513258  0.58395817]
[ 0.11134522 -0.80415605]
[-0.96709276  1.53828669]
[-0.67297331  1.39369146]
[-1.26121221  0.49720103]
[-0.28081405  0.03449629]
[-0.08473441  0.00557724]
[-0.28081405 -0.89091319]
[ 0.89566375 -1.35361793]
[-0.28081405  2.2323438 ]
[ 0.99370357  1.97207239]
[-1.1631724   0.46828198]
[-1.26121221  0.26584866]
[ 1.38586284  1.97207239]
[ 1.28782302 -1.35361793]
[-0.28081405 -0.28361322]
[-0.47689368  1.24909623]
[-0.77101313  1.07558195]
[ 0.99370357 -1.06442747]
[ 0.30742485  0.29476771]
[ 0.99370357  0.75747245]
[-0.67297331 -1.49821316]
[-0.67297331  0.03449629]
[ 0.50350449  1.71180097]
[ 2.07214155  0.17909152]
[-1.94749093 -0.74631796]
[-0.18277423  1.39369146]
[ 0.40546467  0.58395817]
[ 0.89566375 -1.1511846 ]
[-1.1631724  -0.775237  ]
[ 0.20938504  0.23692961]
[ 0.79762394 -0.31253226]
[ 2.07214155 -0.80415605]
[ 0.79762394  0.12125343]
[-0.28081405  0.61287722]
[-0.96709276 -0.31253226]
[ 0.20938504 -0.37037036]
[ 2.07214155  2.11666762]
[ 1.87606192 -1.26686079]
```

```
[ 1.38586284 -0.91983223]
[ 0.89566375  1.24909623]
[ 1.48390265  2.11666762]
[-0.28081405 -1.23794174]
[ 1.97410174  0.90206768]
[ 0.69958412 -0.71739891]
[-1.45729185  0.3526058 ]
[ 0.79762394 -1.35361793]
[ 0.40546467 -0.13901799]
[-0.96709276  0.41044389]
[ 0.0133054  -0.31253226]
[-1.1631724   0.41044389]
[-0.86905295 -1.2090227 ]
[-0.08473441  0.03449629]
[-1.55533166 -0.42820845]
[ 0.99370357 -1.00658937]
[ 1.09174339 -1.2090227 ]
[ 0.0133054  -0.13901799]
[-1.06513258 -1.52713221]
[ 0.79762394 -1.2090227 ]
[ 0.99370357  2.05882953]
[-1.1631724  -1.52713221]
[-0.28081405  0.78639149]
[ 0.11134522 -0.31253226]
[-1.35925203 -1.23794174]
[-0.5749335  -1.49821316]
[ 0.79762394  0.52612008]
[-0.28081405 -0.34145131]
[ 1.7780221  -0.28361322]
[ 0.89566375 -1.03550842]
[ 0.20938504  0.06341534]
[-0.5749335   0.87314863]
[-1.84945111 -1.41145602]
[-1.26121221  0.58395817]
[-0.28081405  0.52612008]
[-0.96709276 -1.09334651]
[ 1.1897832  -1.44037507]
[ 0.20938504 -0.31253226]
[ 1.1897832  -0.74631796]
[-0.28081405  0.06341534]
[ 0.20938504  2.08774857]
[ 0.79762394 -1.09334651]
```

```
[ 0.11134522  0.03449629]
[-1.7514113   0.12125343]
[-0.86905295  0.15017248]
[-0.67297331  0.17909152]
[ 0.89566375 -1.29577984]
[ 0.20938504 -0.25469417]
[-0.37885386  1.22017719]
[ 0.0133054   0.29476771]
[ 0.40546467  0.15017248]
[ 0.89566375 -0.65956082]
[ 0.11134522  0.15017248]
[-1.84945111 -1.29577984]
[-0.08473441  0.29476771]
[-0.18277423 -0.28361322]
[ 0.30742485 -0.51496559]
[-0.18277423  1.59612479]
[ 0.99370357 -1.18010365]
[-0.18277423  1.62504383]
[ 1.28782302  1.8563962 ]
[-1.06513258 -0.37037036]
[ 0.0133054   0.03449629]
[ 0.11134522 -0.25469417]
[-1.55533166 -1.23794174]
[-0.47689368 -0.28361322]
[ 0.99370357  0.12125343]
[ 1.97410174 -1.35361793]
[ 1.48390265  0.06341534]
[-0.5749335   1.36477242]
[ 1.58194247  0.00557724]
[-0.77101313  0.29476771]
[ 1.97410174  0.7285534 ]
[-1.1631724  -0.51496559]
[ 0.69958412  0.26584866]
[-1.35925203 -0.42820845]
[ 0.20938504  0.15017248]
[-0.47689368 -1.2090227 ]
[ 0.6015443   2.00099143]
[-1.55533166 -1.49821316]
[-0.47689368 -0.54388463]
[ 0.50350449  1.82747716]
[-1.35925203 -1.09334651]
[ 0.79762394 -1.38253697]
```

```
[-0.28081405 -0.42820845]
[ 1.58194247  0.98882482]
[ 0.99370357  1.42261051]
[-0.28081405 -0.48604654]
[-0.08473441  2.14558666]
[-1.45729185 -0.11009894]
[-0.08473441  1.94315334]
[-0.67297331 -0.34145131]
[-0.47689368 -0.8330751 ]
[ 0.69958412 -1.38253697]
[-0.77101313 -1.5849703 ]
[-1.84945111 -1.46929411]
[ 1.09174339  0.12125343]
[ 0.11134522  1.50936765]
[-0.28081405  0.09233438]
[ 0.11134522  0.03449629]
[-1.35925203 -1.35361793]
[ 0.30742485  0.06341534]
[-0.86905295  0.38152485]
[ 1.58194247 -1.26686079]
[-0.28081405 -0.74631796]
[-0.08473441  0.15017248]
[-0.86905295 -0.65956082]
[-0.67297331 -0.05226085]
[ 0.40546467 -0.45712749]
[-0.77101313  1.88531525]
[ 1.38586284  1.27801528]
[ 1.1897832  -0.97767033]
[ 1.7780221   1.82747716]
[-0.86905295 -0.25469417]
[-0.77101313  0.55503912]
[-1.1631724  -1.55605125]
[-0.47689368 -1.12226556]
[ 0.30742485  0.06341534]
[-0.18277423 -1.06442747]
[ 1.67998229  1.59612479]
[ 0.99370357  1.76963906]
[ 0.30742485  0.03449629]
[-0.77101313 -0.22577513]
[-0.08473441  0.06341534]
[ 0.30742485 -0.19685608]
[ 1.97410174 -0.65956082]
```

```
[-0.77101313  1.33585337]
[-1.7514113  -0.60172273]
[-0.08473441  0.12125343]
[ 0.30742485 -0.31253226]
[ 1.09174339  0.55503912]
[-0.96709276  0.26584866]
[ 1.48390265  0.3526058 ]
[ 0.20938504 -0.37037036]
[ 2.17018137 -1.03550842]
[-0.28081405  1.104501  ]
[-1.65337148  0.06341534]
[ 0.0133054   0.03449629]
[ 0.11134522  1.04666291]
[-0.08473441 -0.37037036]
[-1.1631724   0.06341534]
[-0.28081405 -1.35361793]
[ 1.58194247  1.104501  ]
[-0.77101313 -1.52713221]
[ 0.11134522  1.8563962 ]
[-0.86905295 -0.775237  ]
[-0.47689368 -0.775237  ]
[-0.28081405 -0.91983223]
[ 0.30742485 -0.71739891]
[ 0.30742485  0.06341534]
[ 0.11134522  1.8563962 ]
[-1.06513258  1.94315334]
[-1.65337148 -1.55605125]
[-1.1631724  -1.09334651]
[-0.67297331 -0.11009894]
[ 0.11134522  0.09233438]
[ 0.30742485  0.26584866]
[ 0.89566375 -0.57280368]
[ 0.30742485 -1.1511846 ]
[-0.08473441  0.67071531]
[ 2.17018137 -0.68847986]
[-1.26121221 -1.38253697]
[-0.96709276 -0.94875128]
[ 0.0133054  -0.42820845]
[-0.18277423 -0.45712749]
[-1.7514113  -0.97767033]
[ 1.7780221   0.98882482]
[ 0.20938504 -0.37037036]
```

```
 [ 0.40546467  1.104501  ]
 [-1.7514113  -1.35361793]
 [ 0.20938504 -0.13901799]
 [ 0.89566375 -1.44037507]
 [-1.94749093  0.46828198]
 [-0.28081405  0.26584866]
 [ 1.87606192 -1.06442747]
 [-0.37885386  0.06341534]
 [ 1.09174339 -0.89091319]
 [-1.06513258 -1.12226556]
 [-1.84945111  0.00557724]
 [ 0.11134522  0.26584866]
 [-1.1631724   0.32368675]
 [-1.26121221  0.29476771]
 [-0.96709276  0.43936294]
 [ 1.67998229 -0.89091319]
 [ 1.1897832   0.52612008]
 [ 1.09174339  0.52612008]
 [ 1.38586284  2.31910094]
 [-0.28081405 -0.13901799]
 [ 0.40546467 -0.45712749]
 [-0.37885386 -0.775237  ]
 [-0.08473441 -0.51496559]
 [ 0.99370357 -1.1511846 ]
 [-0.86905295 -0.775237  ]
 [-0.18277423 -0.51496559]
 [-1.06513258 -0.45712749]
 [-1.1631724   1.39369146]]
```

In [17]:

```
from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(X_train, y_train)
```

Out[17]:

```
▾ GaussianNB
▸ Parameters
```

In [18]:

```
y_pred = classifier.predict(X_test)
```

In [19]:
```
from sklearn.metrics import confusion_matrix, accuracy_score
cm = confusion_matrix(y_test, y_pred)
print(cm)
accuracy_score(y_test, y_pred)
```

```
[[74  5]
 [ 8 33]]
```

Out[19]:
```
0.8916666666666667
```

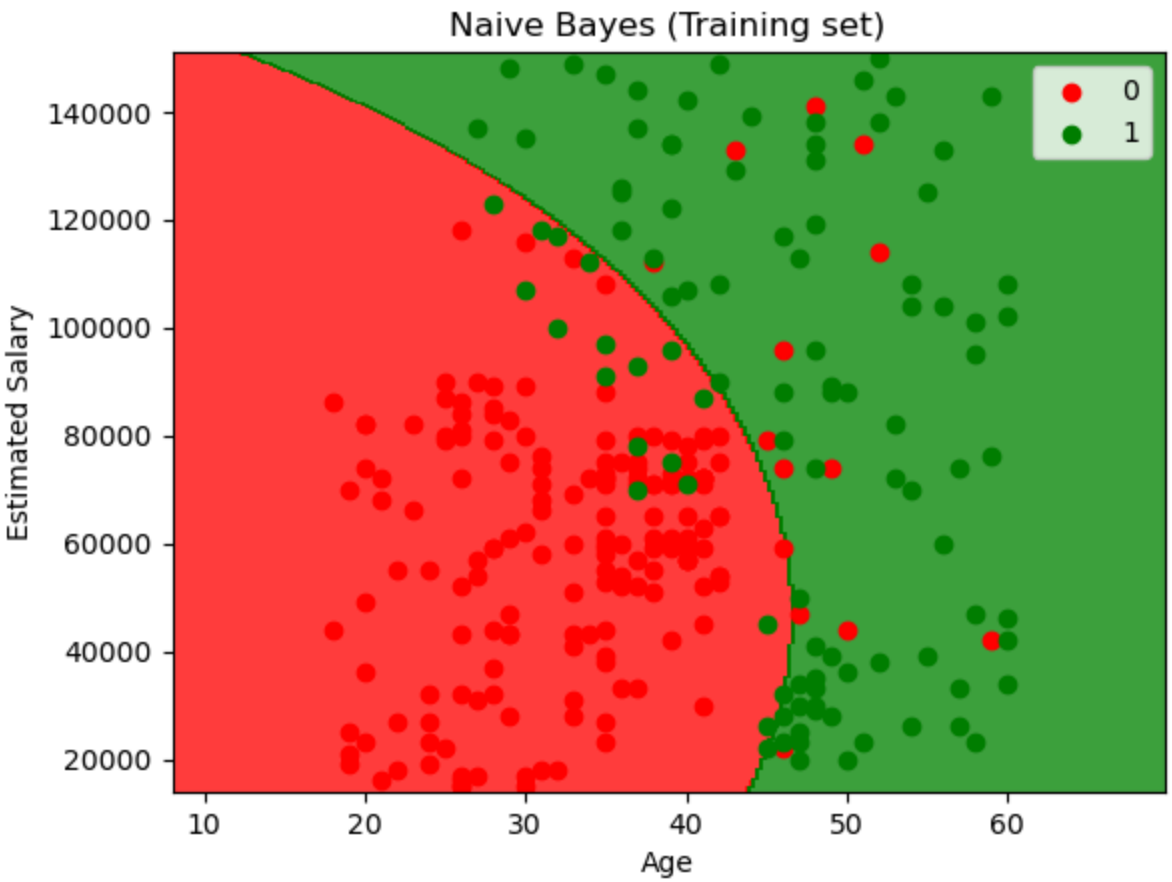
In [20]:
```
from matplotlib.colors import ListedColormap
X_set, y_set = sc.inverse_transform(X_train), y_train
X1, X2 = np.meshgrid(np.arange(start = X_set[:, 0].min() - 10, stop = X_set[:, 0].max() + 10, step = 0.25),
                     np.arange(start = X_set[:, 1].min() - 1000, stop = X_set[:, 1].max() + 1000, step = 0.25))

plt.contourf(X1, X2, classifier.predict(sc.transform(np.array([X1.ravel(), X2.ravel()]).T)).reshape(X1.shape),
             alpha = 0.75, cmap = ListedColormap(('red', 'green')))

plt.xlim(X1.min(), X1.max())
plt.ylim(X2.min(), X2.max())

for i, j in enumerate(np.unique(y_set)):
    plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1],
                color = ListedColormap(('red', 'green'))(i), label = j)

plt.title('Naive Bayes (Training set)')
plt.xlabel('Age')
plt.ylabel('Estimated Salary')
plt.legend()
plt.show()
```



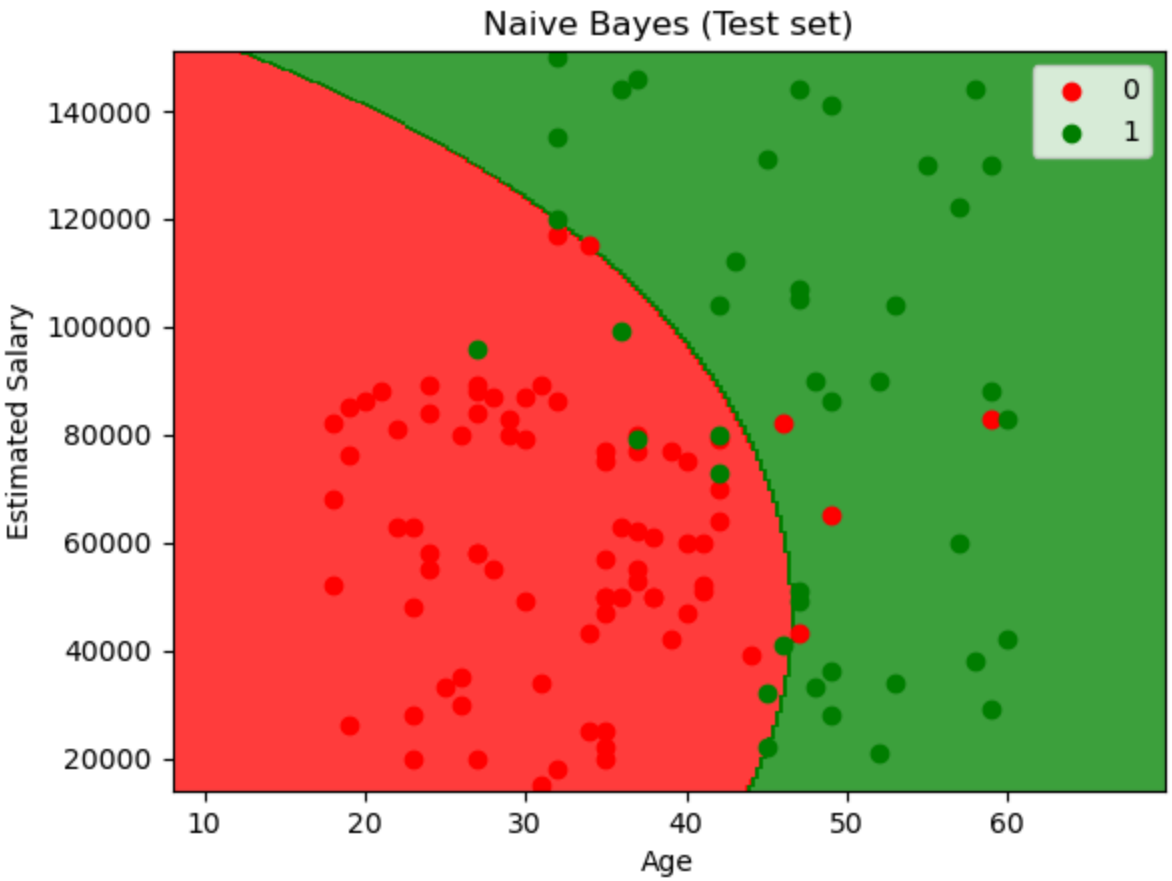
In [21]:
```
from matplotlib.colors import ListedColormap
X_set, y_set = sc.inverse_transform(X_test), y_test
X1, X2 = np.meshgrid(np.arange(start = X_set[:, 0].min() - 10, stop = X_set[:, 0].max() + 10, step = 0.25),
                     np.arange(start = X_set[:, 1].min() - 1000, stop = X_set[:, 1].max() + 1000, step = 0.25))

plt.contourf(X1, X2, classifier.predict(sc.transform(np.array([X1.ravel(), X2.ravel()]).T)).reshape(X1.shape),
             alpha = 0.75, cmap = ListedColormap(('red', 'green')))

plt.xlim(X1.min(), X1.max())
plt.ylim(X2.min(), X2.max())

for i, j in enumerate(np.unique(y_set)):
    plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1],
                color = ListedColormap(('red', 'green'))(i), label = j)
```

```
plt.title('Naive Bayes (Test set)')
plt.xlabel('Age')
plt.ylabel('Estimated Salary')
plt.legend()
plt.show()
```



In [ ]: